

LOCAL

```
piotr@rusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach start --insecure \
  --store=node1 \
  --listen-addr=localhost:26257 \
  --http-addr=localhost:8080 \
  --join=localhost:26257,localhost:26258,localhost:26259 \
  --background
*
* WARNING: ALL SECURITY CONTROLS HAVE BEEN DISABLED!
*
* This mode is intended for non-production testing only.
*
* In this mode:
* - Your cluster is open to any client that can access localhost.
* - Intruders with access to your machine or network can observe client-
  server traffic.
* - Intruders can log in without password and read or write any data in the
  cluster.
* - Intruders can consume all your server's resources and cause
  unavailability.
*
*
* INFO: To start a secure server without mandating TLS for clients,
* consider --accept-sql-without-tls instead. For other options, see:
*
* - https://go.crdb.dev/issue-v/53404/v24.1
* - https://www.cockroachlabs.com/docs/v24.1/secure-a-cluster.html
*
*
* WARNING: Running a server without --sql-addr, with a combined RPC/SQL
  listener, is deprecated.
* This feature will be removed in a later version of CockroachDB.
*
*
* INFO: initial startup completed.
* Node will now attempt to join a running cluster, or wait for `cockroach
  init`.
* Client connections will be accepted after this completes successfully.
* Check the log file(s) for progress.
```

```
piotr@rusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach start --insecure \
  --store=node2 \
  --listen-addr=localhost:26258 \
  --http-addr=localhost:8081 \
  --join=localhost:26257,localhost:26258,localhost:26259 \
```

```

--background
*
* WARNING: ALL SECURITY CONTROLS HAVE BEEN DISABLED!
*
* This mode is intended for non-production testing only.
*
* In this mode:
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* - Intruders with access to your machine or network can observe client-
server traffic.
* - Intruders can log in without password and read or write any data in the
cluster.
* - Intruders can consume all your server's resources and cause
unavailability.
*
*
* INFO: To start a secure server without mandating TLS for clients,
* consider --accept-sql-without-tls instead. For other options, see:
*
* - https://go.crdp.dev/issue-v/53404/v24.1
* - https://www.cockroachlabs.com/docs/v24.1/secure-a-cluster.html
*
*
* WARNING: Running a server without --sql-addr, with a combined RPC/SQL
listener, is deprecated.
* This feature will be removed in a later version of CockroachDB.
*
*
* INFO: initial startup completed.
* Node will now attempt to join a running cluster, or wait for `cockroach
init`.
* Client connections will be accepted after this completes successfully.
* Check the log file(s) for progress.

```

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piotr@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach start --insecure \
  --store=node3 \
  --listen-addr=localhost:26259 \
  --http-addr=localhost:8082 \
  --join=localhost:26257,localhost:26258,localhost:26259 \
  --background
*
* WARNING: ALL SECURITY CONTROLS HAVE BEEN DISABLED!
*
* This mode is intended for non-production testing only.
*
* In this mode:
* - Your cluster is open to any client that can access localhost.
* - Intruders with access to your machine or network can observe client-
server traffic.
* - Intruders can log in without password and read or write any data in the

```

```
cluster.
* - Intruders can consume all your server's resources and cause
unavailability.
*
*
* INFO: To start a secure server without mandating TLS for clients,
* consider --accept-sql-without-tls instead. For other options, see:
*
* - https://go.crdb.dev/issue-v/53404/v24.1
* - https://www.cockroachlabs.com/docs/v24.1/secure-a-cluster.html
*
*
* WARNING: Running a server without --sql-addr, with a combined RPC/SQL
listener, is deprecated.
* This feature will be removed in a later version of CockroachDB.
*
*
* INFO: initial startup completed.
* Node will now attempt to join a running cluster, or wait for `cockroach
init`.
* Client connections will be accepted after this completes successfully.
* Check the log file(s) for progress.
```

```
piotrprusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach init --insecure --host=localhost:26257
Cluster successfully initialized
```

```
piotrprusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach node status --insecure --host=localhost:26257
  id |      address      | sql_address | build |
started_at          |              | updated_at          | locality |
is_available | is_live
-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----
  1 | localhost:26257 | localhost:26257 | v24.1.0 | 2026-04-25
16:37:47.143047 +0000 UTC | 2026-04-25 16:37:56.162934 +0000 UTC |
| true          | true
  2 | localhost:26259 | localhost:26259 | v24.1.0 | 2026-04-25
16:37:47.284401 +0000 UTC | 2026-04-25 16:37:56.294992 +0000 UTC |
| true          | true
  3 | localhost:26258 | localhost:26258 | v24.1.0 | 2026-04-25
16:37:48.20524 +0000 UTC | 2026-04-25 16:37:57.221374 +0000 UTC |
| true          | true
(3 rows)
```

```
piotrprusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach workload init bank 'postgresql://root@localhost:26257?
sslmode=disable' --drop
I260425 16:49:27.315495 1 workload/cli/run.go:639 [-] 1 random seed:
2480145024524052267
I260425 16:49:27.336919 1 ccl/workloadccl/fixture.go:315 [-] 2 starting
import of 1 tables
I260425 16:49:27.646221 35 ccl/workloadccl/fixture.go:492 [-] 3 imported
121 KiB in bank table (1000 rows, 0 index entries, took 275.062871ms, 0.43
MiB/s)
I260425 16:49:27.646299 1 ccl/workloadccl/fixture.go:323 [-] 4 imported
121 KiB bytes in 1 tables (took 309.319444ms, 0.38 MiB/s)
I260425 16:49:27.669880 1 workload/workloadsqli/workloadsqli.go:148 [-] 5
starting 9 splits
```

With three active nodes and initialised bank, I ran a benchmark:

```
piotrprusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach workload run bank 'postgresql://root@localhost:26257?
sslmode=disable' --duration=2m
I260425 16:49:34.202769 1 workload/cli/run.go:639 [-] 1 random seed:
12728946649237182712
I260425 16:49:34.202884 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260425 16:49:34.212618 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 9.733297ms)
_elapsed__errors__ops/sec(inst)__ops/sec(cum)__p50(ms)__p95(ms)__p99(ms)_
pMax(ms)
1.0s 0 656.2 662.7 8.9 24.1 771.8
1006.6 transfer
2.0s 0 977.3 820.0 17.8 54.5 96.5
1677.7 transfer
3.0s 0 1283.4 974.5 12.6 46.1 142.6
486.5 transfer
4.0s 0 1399.5 1080.7 12.6 39.8 151.0
260.0 transfer
5.0s 0 1445.1 1153.6 14.2 37.7 60.8
218.1 transfer
6.0s 0 1514.2 1213.7 13.6 33.6 54.5
75.5 transfer
7.0s 0 1445.8 1246.8 13.1 39.8 88.1
192.9 transfer
8.0s 0 1358.4 1260.8 14.2 46.1 67.1
209.7 transfer
9.0s 0 1433.5 1280.0 14.2 39.8 58.7
104.9 transfer
10.0s 0 1351.8 1287.2 14.7 39.8 67.1
104.9 transfer
11.0s 0 1404.2 1297.8 14.2 35.7 62.9
251.7 transfer
12.0s 0 1761.2 1336.4 11.5 30.4 46.1
```

218.1 transfer						
13.0s	0	1791.9	1371.5	11.5	29.4	41.9
96.5 transfer						
14.0s	0	1696.8	1394.7	11.5	35.7	54.5
104.9 transfer						
15.0s	0	1672.8	1413.2	11.0	35.7	60.8
209.7 transfer						
16.0s	0	1696.9	1431.0	11.0	31.5	60.8
302.0 transfer						
17.0s	0	1715.4	1447.7	12.1	32.5	48.2
159.4 transfer						
18.0s	0	1470.7	1449.0	13.6	32.5	52.4
104.9 transfer						
19.0s	0	1458.0	1449.4	13.6	37.7	56.6
100.7 transfer						
20.0s	0	1457.3	1449.8	13.6	35.7	60.8
88.1 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
21.0s	0	1434.1	1449.1	14.2	37.7	62.9
184.5 transfer						
22.0s	0	1524.7	1452.5	13.1	35.7	50.3
71.3 transfer						
23.0s	0	1593.1	1458.6	13.1	32.5	52.4
79.7 transfer						
24.0s	0	1405.6	1456.4	14.2	37.7	56.6
83.9 transfer						
25.0s	0	1481.8	1457.4	13.6	37.7	54.5
79.7 transfer						
26.0s	0	1556.8	1461.3	12.6	35.7	56.6
83.9 transfer						
27.0s	0	1290.1	1454.9	16.3	39.8	65.0
117.4 transfer						
28.0s	0	1140.6	1443.6	18.9	41.9	71.3
109.1 transfer						
29.0s	0	1445.2	1443.7	14.7	35.7	54.5
104.9 transfer						
30.0s	0	1581.5	1448.3	13.1	32.5	52.4
83.9 transfer						
31.0s	0	1356.7	1445.3	13.6	44.0	67.1
159.4 transfer						
32.0s	0	1124.9	1435.3	18.9	48.2	71.3
151.0 transfer						
33.0s	0	1440.4	1435.5	14.2	37.7	58.7
75.5 transfer						
34.0s	0	1507.8	1437.6	14.2	35.7	52.4
79.7 transfer						
35.0s	0	1456.6	1438.1	13.6	35.7	50.3
96.5 transfer						
36.0s	0	1439.2	1438.2	14.2	35.7	52.4
79.7 transfer						
37.0s	0	1446.2	1438.4	14.2	35.7	50.3
67.1 transfer						
38.0s	0	1533.0	1440.9	13.6	31.5	52.4

71.3 transfer						
39.0s	0	1332.5	1438.1	15.7	39.8	56.6
92.3 transfer						
40.0s	0	1309.0	1434.9	15.2	37.7	65.0
419.4 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
41.0s	0	1304.9	1431.7	16.3	39.8	56.6
71.3 transfer						
42.0s	0	1330.1	1429.3	15.7	39.8	58.7
104.9 transfer						
43.0s	0	1176.6	1423.4	16.8	44.0	62.9
92.3 transfer						
44.0s	0	1243.7	1419.3	16.8	39.8	67.1
318.8 transfer						
45.0s	0	1260.4	1415.8	15.2	48.2	83.9
335.5 transfer						
46.0s	0	1326.9	1413.9	15.7	39.8	67.1
159.4 transfer						
47.0s	0	1320.6	1411.9	15.7	37.7	56.6
100.7 transfer						
48.0s	0	1190.0	1407.2	17.8	41.9	60.8
83.9 transfer						
49.0s	0	1300.3	1405.1	16.8	35.7	54.5
88.1 transfer						
50.0s	0	1215.9	1401.3	17.8	39.8	60.8
92.3 transfer						
51.0s	0	1144.0	1396.2	17.8	50.3	75.5
121.6 transfer						
52.0s	0	1314.3	1394.7	15.7	41.9	56.6
88.1 transfer						
53.0s	0	1416.5	1395.1	13.6	39.8	79.7
125.8 transfer						
54.0s	0	1377.9	1394.8	15.2	37.7	56.6
109.1 transfer						
55.0s	0	1271.5	1392.5	16.8	39.8	56.6
117.4 transfer						
56.0s	0	1237.6	1389.7	16.3	46.1	67.1
96.5 transfer						
57.0s	0	1240.0	1387.1	17.8	39.8	58.7
92.3 transfer						
58.0s	0	1197.4	1383.8	16.8	46.1	67.1
125.8 transfer						
59.0s	0	1216.0	1381.0	17.8	39.8	60.8
104.9 transfer						
60.0s	0	1099.8	1376.3	19.9	44.0	62.9
92.3 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
61.0s	0	1095.3	1371.7	19.9	46.1	60.8
92.3 transfer						
62.0s	0	1005.5	1365.8	21.0	48.2	83.9
109.1 transfer						
63.0s	0	948.0	1359.2	22.0	52.4	88.1

121.6 transfer						
64.0s	0	942.6	1352.7	23.1	48.2	71.3
117.4 transfer						
65.0s	0	855.4	1345.0	25.2	58.7	96.5
184.5 transfer						
66.0s	0	936.4	1338.8	22.0	54.5	83.9
121.6 transfer						
67.0s	0	899.1	1332.3	23.1	54.5	79.7
302.0 transfer						
68.0s	0	928.7	1326.3	23.1	58.7	92.3
151.0 transfer						
69.0s	0	1075.6	1322.7	19.9	48.2	79.7
142.6 transfer						
70.0s	0	1080.1	1319.2	19.9	46.1	75.5
167.8 transfer						
71.0s	0	1025.3	1315.1	18.9	48.2	83.9
385.9 transfer						
72.0s	0	873.1	1308.9	25.2	52.4	75.5
104.9 transfer						
73.0s	0	1036.3	1305.2	21.0	48.2	65.0
75.5 transfer						
74.0s	0	969.1	1300.7	22.0	54.5	79.7
151.0 transfer						
75.0s	0	1129.1	1298.4	17.8	54.5	71.3
100.7 transfer						
76.0s	0	1276.7	1298.1	16.3	41.9	54.5
109.1 transfer						
77.0s	0	1221.2	1297.1	16.8	44.0	62.9
184.5 transfer						
78.0s	0	1141.8	1295.1	17.8	48.2	67.1
100.7 transfer						
79.0s	0	1091.3	1292.5	17.8	52.4	79.7
125.8 transfer						
80.0s	0	1072.0	1289.8	19.9	48.2	71.3
142.6 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
81.0s	0	1087.8	1287.3	18.9	46.1	71.3
121.6 transfer						
82.0s	0	1006.0	1283.9	19.9	50.3	83.9
142.6 transfer						
83.0s	0	949.8	1279.8	23.1	54.5	75.5
130.0 transfer						
84.0s	0	843.5	1274.6	25.2	62.9	104.9
251.7 transfer						
85.0s	0	969.5	1271.0	21.0	52.4	71.3
109.1 transfer						
86.0s	0	919.1	1267.0	22.0	52.4	79.7
130.0 transfer						
87.0s	0	1040.0	1264.3	21.0	54.5	75.5
159.4 transfer						
88.0s	0	1036.5	1261.8	19.9	52.4	75.5
125.8 transfer						
89.0s	0	1302.7	1262.2	15.7	39.8	58.7

83.9 transfer						
90.0s	0	1202.2	1261.5	17.8	44.0	65.0
75.5 transfer						
91.0s	0	1301.9	1262.0	15.2	41.9	71.3
226.5 transfer						
92.0s	0	1363.4	1263.1	15.2	37.7	62.9
109.1 transfer						
93.0s	0	1317.9	1263.7	15.7	41.9	60.8
96.5 transfer						
94.0s	0	1309.8	1264.2	15.7	39.8	62.9
130.0 transfer						
95.0s	0	1336.6	1264.9	15.7	39.8	58.7
75.5 transfer						
96.0s	0	1318.9	1265.5	15.2	41.9	65.0
100.7 transfer						
97.0s	0	1287.5	1265.7	15.7	44.0	71.3
130.0 transfer						
98.0s	0	1311.3	1266.2	16.3	41.9	60.8
83.9 transfer						
99.0s	0	1307.3	1266.6	15.7	39.8	58.7
88.1 transfer						
100.0s	0	1302.1	1267.0	16.3	39.8	60.8
83.9 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
101.0s	0	1267.7	1267.0	16.3	39.8	62.9
100.7 transfer						
102.0s	0	1276.5	1267.1	16.3	41.9	58.7
100.7 transfer						
103.0s	0	1305.3	1267.4	16.3	39.8	58.7
117.4 transfer						
104.0s	0	1266.7	1267.4	16.8	39.8	54.5
83.9 transfer						
105.0s	0	1302.9	1267.8	16.3	37.7	60.8
92.3 transfer						
106.0s	0	1180.4	1266.9	16.8	46.1	79.7
151.0 transfer						
107.0s	0	1340.4	1267.6	15.2	39.8	62.9
96.5 transfer						
108.0s	0	1355.8	1268.4	15.7	39.8	52.4
92.3 transfer						
109.0s	0	1221.7	1268.0	16.8	44.0	71.3
318.8 transfer						
110.0s	0	1282.6	1268.1	16.3	41.9	58.7
125.8 transfer						
111.0s	0	1347.9	1268.9	15.2	39.8	65.0
218.1 transfer						
112.0s	0	1358.2	1269.7	15.7	35.7	56.6
75.5 transfer						
113.0s	0	1366.6	1270.5	15.2	39.8	54.5
100.7 transfer						
114.0s	0	1373.3	1271.4	15.2	39.8	56.6
79.7 transfer						
115.0s	0	1343.9	1272.1	15.7	37.7	58.7


```

100.7 transfer
 116.0s      0      1347.1      1272.7      15.2      41.9      58.7
79.7 transfer
 117.0s      0      1225.3      1272.3      15.7      50.3      79.7
130.0 transfer
 118.0s      0      1293.9      1272.5      15.7      39.8      60.8
88.1 transfer
 119.0s      0      1262.9      1272.4      16.3      39.8      75.5
109.1 transfer
 120.0s      0      1289.9      1272.5      15.7      44.0      65.0
96.5 transfer

_elapsed__errors____ops(total)____ops/sec(cum)____avg(ms)____p50(ms)____p95(ms)____
_p99(ms)_pMax(ms)____total
 120.0s      0      152706      1272.5      18.9      15.7      41.9
67.1  1677.7 transfer

_elapsed__errors____ops(total)____ops/sec(cum)____avg(ms)____p50(ms)____p95(ms)____
_p99(ms)_pMax(ms)____result
 120.0s      0      152706      1272.5      18.9      15.7      41.9
67.1  1677.7

```

After full benchmark, I tested what if second node dies while benchmarking:

```

piotr@rusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach workload run bank 'postgresql://root@localhost:26257?
sslmode=disable' --duration=5m
I260425 16:54:22.791269 1 workload/cli/run.go:639 [-] 1 random seed:
11146626805739849253
I260425 16:54:22.791336 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260425 16:54:22.792930 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 1.592697ms)
_elapsed__errors__ops/sec(inst)____ops/sec(cum)____p50(ms)____p95(ms)____p99(ms)____
pMax(ms)
 1.0s      0      442.0      442.8      7.9      17.8      771.8
906.0 transfer
 2.0s      0      1321.2      882.0      11.5      32.5      67.1
1744.8 transfer
 3.0s      0      1480.3      1081.3      13.1      37.7      75.5
2080.4 transfer
 4.0s      0      1468.3      1178.0      13.1      37.7      65.0
100.7 transfer
 5.0s      0      1472.2      1236.9      13.6      35.7      60.8
134.2 transfer
 6.0s      0      1470.3      1275.7      13.6      37.7      60.8
117.4 transfer
 7.0s      0      1428.5      1297.6      12.1      37.7      142.6
402.7 transfer
 8.0s      0      1465.9      1318.6      14.2      35.7      54.5
130.0 transfer
 9.0s      0      1364.3      1323.7      15.2      41.9      67.1

```

142.6 transfer						
10.0s	0	1348.4	1326.1	14.7	39.8	71.3
201.3 transfer						
11.0s	0	1219.4	1316.4	15.2	41.9	121.6
285.2 transfer						
12.0s	0	1413.8	1324.6	13.6	39.8	88.1
268.4 transfer						
13.0s	0	1388.0	1329.4	14.2	41.9	65.0
88.1 transfer						
14.0s	0	1439.5	1337.3	14.7	37.7	52.4
79.7 transfer						
15.0s	0	1445.6	1344.5	14.2	37.7	62.9
83.9 transfer						
16.0s	0	1383.1	1346.9	14.7	37.7	71.3
218.1 transfer						
17.0s	0	1389.9	1349.5	14.7	39.8	71.3
100.7 transfer						
18.0s	0	1366.6	1350.4	14.7	41.9	65.0
92.3 transfer						
19.0s	0	1383.9	1352.2	14.7	39.8	62.9
92.3 transfer						
20.0s	0	1272.7	1348.2	15.2	46.1	71.3
121.6 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
21.0s	0	1336.6	1347.7	14.7	37.7	67.1
385.9 transfer						
22.0s	0	1341.1	1347.4	15.7	37.7	56.6
96.5 transfer						
23.0s	0	1295.3	1345.1	16.3	37.7	60.8
92.3 transfer						
24.0s	0	1369.6	1346.1	14.7	41.9	60.8
88.1 transfer						
25.0s	0	1437.9	1349.8	14.2	37.7	56.6
88.1 transfer						
26.0s	0	1159.3	1342.5	17.8	46.1	71.3
100.7 transfer						
27.0s	0	1332.6	1342.1	15.2	39.8	60.8
92.3 transfer						
28.0s	0	1398.7	1344.1	13.6	44.0	67.1
335.5 transfer						
29.0s	0	1415.1	1346.6	13.6	37.7	62.9
142.6 transfer						
30.0s	0	1327.4	1345.9	12.1	33.6	121.6
906.0 transfer						
31.0s	0	443.3	1316.8	14.7	35.7	117.4
805.3 transfer						
32.0s	0	0.0	1275.6	0.0	0.0	0.0
0.0 transfer						
33.0s	0	0.0	1237.0	0.0	0.0	0.0
0.0 transfer						
34.0s	0	0.0	1200.6	0.0	0.0	0.0
0.0 transfer						
35.0s	0	0.0	1166.3	0.0	0.0	0.0

```

0.0 transfer
  36.0s      0      0.0      1133.9      0.0      0.0      0.0
0.0 transfer
Error: pq: result is ambiguous: error=ba: Put [/Table/106/1/400/0],
EndTxn(parallel commit) [/Table/106/1/400/0], [txn: 2cce471c] RPC error:
grpc: error reading from server: read tcp 127.0.0.1:54952->127.0.0.1:26258:
read: connection reset by peer [code 14/Unavailable] [exhausted] (last
error: routing information detected to be stale: [NotLeaseHolderError]
lease held by different store; r73: replica (n3,s3):2 not lease holder;
current lease is repl=(n2,s2):3 seq=3 start=1777135826.439980668,0 epo=1
pro=1777135826.450045745,0)

```

After killing node2 during the workload, throughput dropped from ~1350 ops/sec to 0 ops/sec for several seconds. This temporary unavailability is caused by Raft leader election and leaseholder transfer. The client observed an ambiguous result error caused by connection reset and leaseholder change. This shows that CockroachDB prioritizes consistency: instead of returning possibly incorrect results, it reports uncertainty and requires the client to retry.

After that test I revived node2:

```

piotr@rusak@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach start --insecure \
  --store=node2 \
  --listen-addr=localhost:26258 \
  --http-addr=localhost:8081 \
  --join=localhost:26257,localhost:26258,localhost:26259 \
  --background
*
* WARNING: ALL SECURITY CONTROLS HAVE BEEN DISABLED!
*
* This mode is intended for non-production testing only.
*
* In this mode:
* - Your cluster is open to any client that can access localhost.
* - Intruders with access to your machine or network can observe client-
  server traffic.
* - Intruders can log in without password and read or write any data in the
  cluster.
* - Intruders can consume all your server's resources and cause
  unavailability.
*
*
* INFO: To start a secure server without mandating TLS for clients,
* consider --accept-sql-without-tls instead. For other options, see:
*
* - https://go.crdb.dev/issue-v/53404/v24.1
* - https://www.cockroachlabs.com/docs/v24.1/secure-a-cluster.html
*
*
* WARNING: Running a server without --sql-addr, with a combined RPC/SQL
  listener, is deprecated.

```

*

-----+-----+-----+-----+-----
-----+-----+-----+-----+-----
-----+-----

```

piotr@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach workload run bank 'postgresql://root@localhost:26257?
sslmode=disable' --duration=5m
I260425 17:42:14.560950 1 workload/cli/run.go:639 [-] 1 random seed:
13914126934278834086
I260425 17:42:14.561035 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260425 17:42:14.562258 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 1.222959ms)
_elapsed__errors__ops/sec(inst)__ops/sec(cum)__p50(ms)__p95(ms)__p99(ms)__
pMax(ms)
1.0s 0 545.0 545.8 8.1 17.8 838.9
973.1 transfer
2.0s 0 1079.1 812.4 12.1 37.7 83.9
1744.8 transfer
3.0s 0 1364.0 996.3 13.1 35.7 125.8
2415.9 transfer
4.0s 0 1254.3 1060.9 16.3 41.9 75.5
104.9 transfer
5.0s 0 1284.8 1105.5 16.3 39.8 56.6
92.3 transfer
6.0s 0 1156.5 1114.1 17.8 44.0 62.9
151.0 transfer
7.0s 0 1251.6 1133.7 14.2 44.0 88.1
469.8 transfer

```

8.0s	0	1346.3	1160.3	14.7	41.9	83.9
260.0 transfer						
9.0s	0	1207.3	1165.5	15.2	50.3	75.5
142.6 transfer						
10.0s	0	1220.7	1171.0	15.7	48.2	79.7
117.4 transfer						
11.0s	0	1403.9	1192.2	12.6	41.9	151.0
192.9 transfer						
12.0s	0	1480.0	1216.2	13.6	37.7	56.6
96.5 transfer						
13.0s	0	1432.5	1232.8	14.7	35.7	54.5
67.1 transfer						
14.0s	0	712.2	1195.6	14.2	37.7	54.5
176.2 transfer						
15.0s	0	0.0	1116.0	0.0	0.0	0.0
0.0 transfer						
16.0s	0	0.0	1046.2	0.0	0.0	0.0
0.0 transfer						
17.0s	0	0.0	984.7	0.0	0.0	0.0
0.0 transfer						

```
Error: pq: result is ambiguous: error=result is ambiguous: error=failed to
connect to n3 at localhost:26259: grpc: connection error: desc =
"transport: error while dialing: connection interrupted (did the remote
node shut down or are there networking issues?)" [code 14/Unavailable]
[exhausted] (last error: failed to connect to n3 at localhost:26259: grpc:
connection error: desc = "transport: error while dialing: connection
interrupted (did the remote node shut down or are there networking
issues?)" [code 14/Unavailable]) [exhausted] (last error: result is
ambiguous: error=failed to connect to n3 at localhost:26259: grpc:
connection error: desc = "transport: error while dialing: connection
interrupted (did the remote node shut down or are there networking
issues?)" [code 14/Unavailable] [exhausted] (last error: failed to connect
to n3 at localhost:26259: grpc: connection error: desc = "transport: error
while dialing: connection interrupted (did the remote node shut down or are
there networking issues?)" [code 14/Unavailable]))
```

During node failure, the system experienced a temporary drop in throughput and increased latency. This behavior is caused by Raft leader re-election and leaseholder transfer. After a short recovery period, the system resumed normal operation, demonstrating fault tolerance with strong consistency guarantees.

Then, I wanted to figure out what if error wouldn't terminate the benchmark:

```
piotr@fedora:~/Agh/lsc-db-benchmark/cockroach-v24.1.0.linux-amd64$
./cockroach workload run bank 'postgres://root@localhost:26257?
sslmode=disable' --duration=5m --tolerate-errors
I260425 18:00:01.342215 1 workload/cli/run.go:639 [-] 1 random seed:
6875242170990325666
I260425 18:00:01.342305 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260425 18:00:01.343676 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 1.369974ms)
```

_elapsed	__errors	__ops/sec(inst)	__ops/sec(cum)	__p50(ms)	__p95(ms)	__p99(ms)	_pMax(ms)
1.0s	0	83.8	84.0	7.1	15.2	113.2	
121.6 transfer							
2.0s	0	368.8	226.4	8.9	23.1	1879.0	
2013.3 transfer							
3.0s	0	1166.6	539.6	12.1	79.7	151.0	
2415.9 transfer							
4.0s	0	1146.6	691.4	11.5	83.9	159.4	
243.3 transfer							
5.0s	0	1159.1	785.0	13.1	60.8	159.4	
302.0 transfer							
6.0s	0	1165.7	848.4	12.6	75.5	201.3	
419.4 transfer							
7.0s	0	1063.5	879.1	15.2	71.3	159.4	
260.0 transfer							
8.0s	0	1180.4	916.8	16.8	44.0	100.7	
167.8 transfer							
9.0s	0	1135.6	941.1	15.2	50.3	134.2	
385.9 transfer							
10.0s	0	1070.2	954.0	13.6	44.0	285.2	
352.3 transfer							
11.0s	0	1171.7	973.8	12.1	39.8	234.9	
671.1 transfer							
12.0s	0	1038.9	979.2	14.7	37.7	121.6	
704.6 transfer							
13.0s	0	0.0	903.9	0.0	0.0	0.0	
0.0 transfer							
14.0s	0	0.0	839.3	0.0	0.0	0.0	
0.0 transfer							
15.0s	0	0.0	783.4	0.0	0.0	0.0	
0.0 transfer							
16.0s	0	0.0	734.4	0.0	0.0	0.0	
0.0 transfer							
17.0s	0	20.0	692.4	5368.7	5368.7	5368.7	
5368.7 transfer							
E260425 18:00:18.390183 1 workload/cli/run.go:568 [-] 4 pq: result is ambiguous: error=ba: Put [/Table/106/1/860/0], [txn: 84afc5e8] RPC error: grpc: error reading from server: read tcp 127.0.0.1:45618->127.0.0.1:26259: read: connection reset by peer [code 14/Unavailable] [exhausted] (last error: routing information detected to be stale: [NotLeaseHolderError] lease held by different store; r72: replica (n2,s2):3 not lease holder; current lease is repl=(n3,s3):2 seq=9 start=1777140009.541325632,0 epo=8 pro=1777140009.553806779,0)							
18.0s	2	1380.7	730.7	13.1	39.8	92.3	
5905.6 transfer							
19.0s	2	1482.3	770.2	13.6	35.7	60.8	
125.8 transfer							
20.0s	2	1462.7	804.8	14.2	37.7	54.5	
83.9 transfer							
_elapsed	__errors	__ops/sec(inst)	__ops/sec(cum)	__p50(ms)	__p95(ms)	__p99(ms)	_pMax(ms)
21.0s	2	1478.7	836.9	13.6	37.7	56.6	
125.8 transfer							

22.0s	2	1482.2	866.3	13.6	35.7	58.7
125.8 transfer						
23.0s	2	1502.6	893.9	13.6	35.7	50.3
104.9 transfer						
24.0s	2	1519.4	920.0	14.2	35.7	50.3
62.9 transfer						
25.0s	2	1462.0	941.7	14.7	35.7	50.3
75.5 transfer						
26.0s	2	1475.3	962.3	14.2	35.7	50.3
75.5 transfer						
27.0s	2	1388.0	978.0	14.7	37.7	58.7
109.1 transfer						
28.0s	2	1376.5	992.2	15.7	37.7	56.6
79.7 transfer						
29.0s	2	961.6	991.2	21.0	54.5	79.7
117.4 transfer						
30.0s	2	1294.6	1001.3	15.2	44.0	83.9
130.0 transfer						
31.0s	2	1342.7	1012.3	15.2	41.9	58.7
117.4 transfer						
32.0s	2	1570.6	1029.7	13.1	37.7	54.5
71.3 transfer						
33.0s	2	1520.8	1044.6	13.1	35.7	50.3
251.7 transfer						
34.0s	2	1540.8	1059.2	13.6	35.7	50.3
79.7 transfer						
35.0s	2	1506.5	1072.0	13.6	37.7	52.4
92.3 transfer						
36.0s	2	1515.9	1084.3	13.6	35.7	54.5
92.3 transfer						
37.0s	2	1469.3	1094.7	13.6	39.8	58.7
104.9 transfer						
38.0s	2	1495.5	1105.3	13.6	37.7	52.4
96.5 transfer						
39.0s	2	1424.9	1113.5	14.2	37.7	54.5
113.2 transfer						
40.0s	2	1418.4	1121.1	14.2	39.8	60.8
113.2 transfer						
_elapsed__errors__ops/sec(inst)____ops/sec(cum)____p50(ms)____p95(ms)____p99(ms)____pMax(ms)						
41.0s	2	1448.7	1129.1	14.2	37.7	54.5
100.7 transfer						
42.0s	2	1434.9	1136.4	14.7	35.7	50.3
104.9 transfer						
43.0s	2	1294.6	1140.0	16.3	39.8	56.6
83.9 transfer						
44.0s	2	1224.8	1142.0	16.8	44.0	60.8
121.6 transfer						
45.0s	2	1474.2	1149.4	14.7	35.7	50.3
100.7 transfer						
46.0s	2	1530.5	1157.6	13.1	35.7	52.4
192.9 transfer						
47.0s	2	1502.6	1165.0	13.1	35.7	50.3
71.3 transfer						

48.0s	2	1501.5	1172.0	13.6	39.8	58.7
109.1 transfer						
49.0s	2	1528.7	1179.3	13.6	35.7	48.2
75.5 transfer						
50.0s	2	1561.1	1186.9	13.6	30.4	50.3
209.7 transfer						
51.0s	2	1580.9	1194.6	13.6	32.5	46.1
60.8 transfer						
52.0s	2	1553.6	1201.5	13.1	35.7	52.4
83.9 transfer						
53.0s	2	1519.7	1207.5	13.6	35.7	48.2
71.3 transfer						
54.0s	2	1508.5	1213.1	13.6	35.7	52.4
159.4 transfer						
55.0s	2	1559.1	1219.4	13.1	35.7	58.7
88.1 transfer						
56.0s	2	1474.0	1223.9	13.1	39.8	67.1
268.4 transfer						
57.0s	2	1419.7	1227.4	14.2	37.7	71.3
151.0 transfer						
58.0s	2	1298.3	1228.6	15.7	44.0	60.8
96.5 transfer						
59.0s	2	1239.8	1228.8	16.8	44.0	62.9
192.9 transfer						
60.0s	2	1256.2	1229.2	16.3	41.9	65.0
268.4 transfer						
_elapsed__errors__ops/sec(inst)__ops/sec(cum)__p50(ms)__p95(ms)__p99(ms)_pMax(ms)						

As we can see, ops/sec briefly dropped to 0 and then recovered to previous values.

AWS

Configuring AWS required 3 EC2 instances and security group that enables inbound tcp on 26257. Each node was deployed on a separate EC2 instance within the same VPC to enable low-latency communication.

EC2

Dashboard

AWS Global View

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Instances

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Spot Requests

Savings Plans

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Auto Scaling Groups

Instances (3)

Info

Connect

Instance state

Actions

Launch Instances

Save filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	cockdb-bench...	i-0da86bd6dd1c0acaf	Running	t3.micro	Initializing	View alarms +	us-east-1d	ec2-35-173-241-110.co...	35.173.241.110	-	-
☐	cockdb-bench...	i-071cf981bd377c49	Running	t3.micro	Initializing	View alarms +	us-east-1d	ec2-34-207-132-67.co...	34.207.132.67	-	-
☐	cockdb-bench...	i-0b2d5dee72794d923	Running	t3.micro	Initializing	View alarms +	us-east-1d	ec2-54-224-157-90.co...	54.224.157.90	-	-

Select an instance

EC2

Dashboard

AWS Global View

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Instances

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Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Security Groups

sg-022361798e19b38be - launch-wizard-1

Inbound security group rules successfully modified on security group (sg-022361798e19b38be | launch-wizard-1)

Details

sg-022361798e19b38be - launch-wizard-1

Actions

Details

Security group name

launch-wizard-1

Security group ID

sg-022361798e19b38be

Description

launch-wizard-1 created 2026-04-26T14:16:29.977Z

VPC ID

vpc-054686d6eb31b2bee

Owner

471112675803

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-01eaab977d9cf2d53	IPv4	Custom TCP	TCP	26257	172.31.0.0/16	-
☐	-	sgr-0189d827ed7c8dbbd	IPv4	SSH	TCP	22	0.0.0.0/0	-

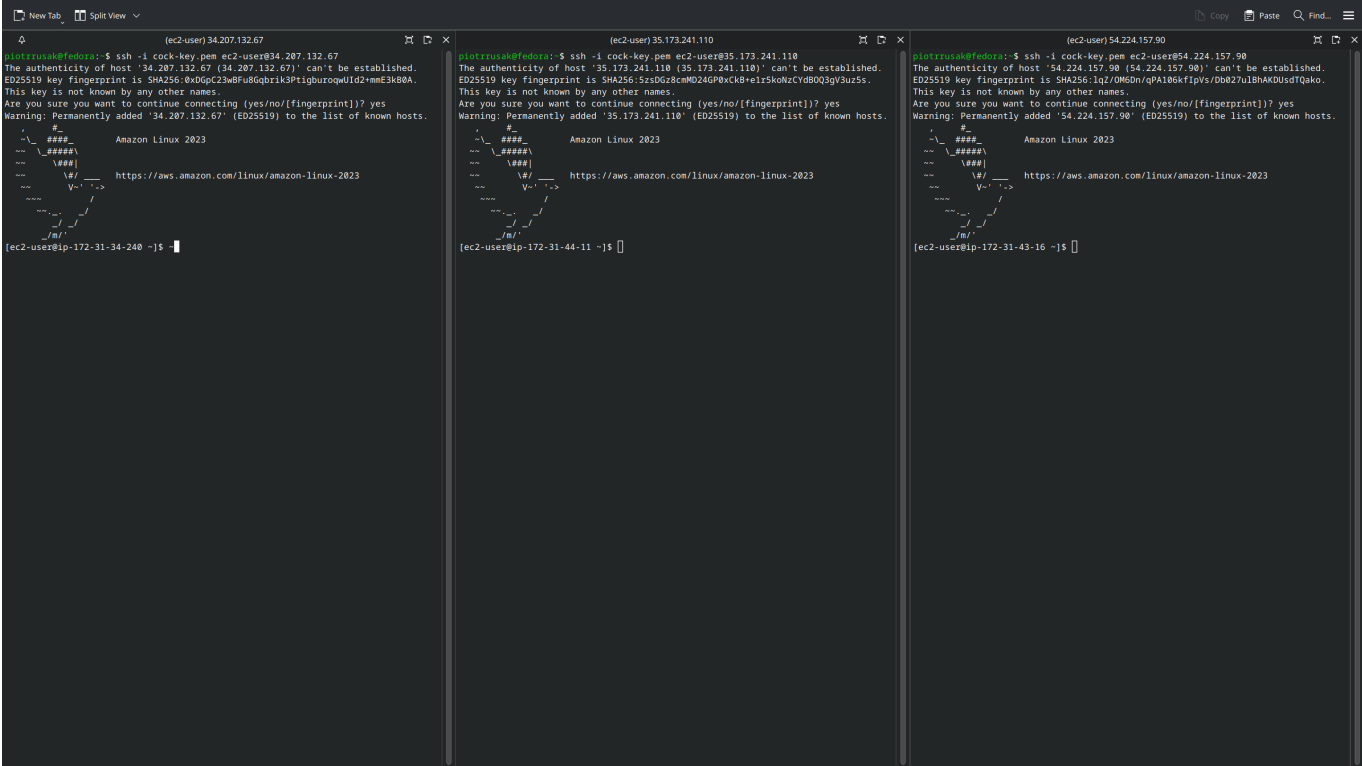
cloudShellFeedback

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cloudShellFeedback

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/



Baseline:

```
[ec2-user@ip-172-31-34-240 cockroach-v24.1.0.linux-amd64]$ ./cockroach
workload run bank 'postgres://root@172.31.34.240:26257?sslmode=disable' -
-duration=5m
I260426 14:59:28.344874 1 workload/cli/run.go:639 [-] 1 random seed:
9359561712330100429
I260426 14:59:28.345126 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260426 14:59:28.780259 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 435.12857ms)
_elapsed__errors__ops/sec(inst)__ops/sec(cum)__p50(ms)__p95(ms)__p99(ms)_
pMax(ms)
1.0s      0      108.3      155.7      16.8      37.7      285.2
369.1 transfer
2.0s      0      138.4      147.3      18.9      71.3      117.4
125.8 transfer
3.0s      0      8.0      100.9      369.1      402.7      402.7
402.7 transfer
4.0s      0      7.6      77.1      385.9      486.5      486.5
486.5 transfer
5.0s      0      4.1      62.9      1073.7      1073.7      1073.7
1073.7 transfer
6.0s      0      4.0      53.1      1073.7      1140.9      1140.9
1140.9 transfer
7.0s      0      2.0      45.8      1006.6      1140.9      1140.9
1140.9 transfer
8.0s      0      2.0      40.4      1811.9      1811.9      1811.9
1811.9 transfer
9.0s      0      0.0      35.9      0.0      0.0      0.0
0.0 transfer
```

10.0s	0	2.0	32.5	2684.4	2818.6	2818.6
2818.6 transfer						
11.0s	0	0.0	29.6	0.0	0.0	0.0
0.0 transfer						
12.0s	0	0.0	27.1	0.0	0.0	0.0
0.0 transfer						
13.0s	0	0.0	25.0	0.0	0.0	0.0
0.0 transfer						
14.0s	0	0.0	23.2	0.0	0.0	0.0
0.0 transfer						
15.0s	0	0.0	21.7	0.0	0.0	0.0
0.0 transfer						
16.0s	0	1.0	20.4	8321.5	8321.5	8321.5
8321.5 transfer						
17.0s	0	0.0	19.2	0.0	0.0	0.0
0.0 transfer						
18.0s	0	0.0	18.1	0.0	0.0	0.0
0.0 transfer						
19.0s	0	1.0	17.2	11811.2	11811.2	11811.2
11811.2 transfer						
20.0s	0	0.0	16.4	0.0	0.0	0.0
0.0 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						

Recovery:

```
[ec2-user@ip-172-31-34-240 cockroach-v24.1.0.linux-amd64]$ ./cockroach
workload run bank 'postgresql://root@172.31.34.240:26257?sslmode=disable' -
-duration=5m
I260426 15:01:36.117335 1 workload/cli/run.go:639 [-] 1 random seed:
3532145109929266603
I260426 15:01:36.446508 1 workload/cli/run.go:431 [-] 2 creating load
generator...
I260426 15:03:40.540083 1 workload/cli/run.go:470 [-] 3 creating load
generator... done (took 2m4.103914139s)
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___
pMax(ms)
 45.0s      0          0.0          0.0 45097.2 45097.2 45097.2
45097.2 transfer
 46.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
 47.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
 48.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
 49.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
 50.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
 51.0s      0          0.0          0.0  0.0  0.0  0.0
0.0 transfer
```

52.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
53.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
54.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
55.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
56.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
57.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
58.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
59.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
60.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
61.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
62.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
63.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
64.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
65.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
66.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
67.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
68.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
69.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
70.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
71.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
72.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
73.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
74.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
75.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
76.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
77.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						

78.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
79.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
80.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
81.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
82.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
83.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
84.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
_elapsed__errors__ops/sec(inst)__ops/sec(cum)__p50(ms)__p95(ms)__p99(ms)__pMax(ms)						
85.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
86.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
87.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
88.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
89.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
90.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
91.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
92.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
93.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
94.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
95.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
96.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
97.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
98.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
99.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
100.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
101.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
102.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
103.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						

104.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
105.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
106.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
107.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
108.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
109.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
110.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
111.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
112.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
113.1s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
114.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
115.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
116.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
117.0s	0	0.0	0.0	0.0	0.0	0.0
0.0 transfer						
118.0s	0	3.0	0.0	103079.2	103079.2	103079.2
103079.2 transfer						
119.0s	0	149.0	1.3	13.1	22.0	1342.2
77309.4 transfer						
120.0s	0	267.8	3.5	13.6	24.1	44.0
56.6 transfer						
121.0s	0	260.2	5.6	14.2	28.3	39.8
50.3 transfer						
122.0s	0	251.0	7.6	14.7	27.3	31.5
41.9 transfer						
123.0s	0	289.8	9.9	13.6	19.9	31.5
41.9 transfer						
124.0s	0	293.0	12.2	13.1	19.9	28.3
41.9 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
125.0s	0	270.0	14.3	13.6	25.2	33.6
35.7 transfer						
126.0s	0	265.0	16.3	14.2	25.2	56.6
65.0 transfer						
127.0s	0	279.9	18.3	14.2	21.0	32.5
46.1 transfer						
128.0s	0	95.0	18.9	14.7	37.7	536.9
536.9 transfer						

129.0s	0	0.0	18.8	0.0	0.0	0.0
0.0 transfer						
130.0s	0	0.0	18.7	0.0	0.0	0.0
0.0 transfer						
131.0s	0	0.0	18.5	0.0	0.0	0.0
0.0 transfer						
132.0s	0	0.0	18.4	0.0	0.0	0.0
0.0 transfer						
133.0s	0	0.0	18.2	0.0	0.0	0.0
0.0 transfer						
134.0s	0	0.0	18.1	0.0	0.0	0.0
0.0 transfer						
135.0s	0	0.0	18.0	0.0	0.0	0.0
0.0 transfer						
136.0s	0	0.0	17.8	0.0	0.0	0.0
0.0 transfer						
137.0s	0	0.0	17.7	0.0	0.0	0.0
0.0 transfer						
138.0s	0	0.0	17.6	0.0	0.0	0.0
0.0 transfer						
139.0s	0	0.0	17.4	0.0	0.0	0.0
0.0 transfer						
140.0s	0	0.0	17.3	0.0	0.0	0.0
0.0 transfer						
141.0s	0	0.0	17.2	0.0	0.0	0.0
0.0 transfer						
142.0s	0	0.0	17.1	0.0	0.0	0.0
0.0 transfer						
143.0s	0	60.1	17.4	41.9	14495.5	15032.4
15032.4 transfer						
144.0s	0	151.8	18.3	19.9	58.7	92.3
100.7 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						
145.0s	0	277.3	20.1	14.7	22.0	26.2
37.7 transfer						
146.0s	0	285.0	21.9	13.6	21.0	30.4
50.3 transfer						
147.0s	0	284.2	23.7	13.6	21.0	29.4
41.9 transfer						
148.0s	0	294.9	25.5	13.1	21.0	27.3
39.8 transfer						
149.0s	0	275.9	27.2	14.2	21.0	27.3
35.7 transfer						
150.0s	0	276.9	28.9	13.6	23.1	32.5
46.1 transfer						
151.0s	0	298.1	30.7	13.1	19.9	25.2
79.7 transfer						
152.0s	0	282.9	32.3	14.2	21.0	29.4
32.5 transfer						
153.0s	0	285.1	34.0	13.6	22.0	27.3
46.1 transfer						
154.0s	0	296.9	35.7	13.1	22.0	25.2
41.9 transfer						

155.0s	0	269.0	37.2	14.2	26.2	41.9
44.0 transfer						
156.0s	0	277.0	38.7	14.2	22.0	29.4
44.0 transfer						
157.0s	0	239.7	40.0	15.2	33.6	46.1
58.7 transfer						
158.0s	0	255.2	41.4	15.2	25.2	41.9
60.8 transfer						
159.0s	0	273.0	42.8	14.2	24.1	37.7
46.1 transfer						
160.0s	0	251.7	44.1	15.2	26.2	33.6
41.9 transfer						
161.0s	0	281.3	45.6	13.6	23.1	29.4
37.7 transfer						
162.0s	0	124.7	46.1	27.3	75.5	109.1
113.2 transfer						
163.0s	0	210.9	47.1	16.3	37.7	88.1
88.1 transfer						
164.0s	0	238.4	48.3	15.7	30.4	46.1
58.7 transfer						
_elapsed__errors__ops/sec(inst)___ops/sec(cum)___p50(ms)___p95(ms)___p99(ms)___pMax(ms)						

Failure & recovery can be easily seen

After recovery test:

```
[ec2-user@ip-172-31-43-16 cockroach-v24.1.0.linux-amd64]$ ./cockroach node
status --insecure --host=172.31.34.240:26257
  id |      address      |      sql_address      | build |
started_at          |          updated_at          | locality |
is_available | is_live
-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+
  1 | 172.31.34.240:26257 | 172.31.34.240:26257 | v24.1.0 | 2026-04-26
14:59:08.400235 +0000 UTC | 2026-04-26 15:10:02.472411 +0000 UTC |
| false | false
  2 | 172.31.43.16:26257 | 172.31.43.16:26257 | v24.1.0 | 2026-04-26
14:48:19.819595 +0000 UTC | 2026-04-26 15:10:01.926362 +0000 UTC |
| false | false
  3 | 172.31.44.11:26257 | 172.31.44.11:26257 | v24.1.0 | 2026-04-26
14:48:20.094994 +0000 UTC | 2026-04-26 15:05:36.868796 +0000 UTC |
| false | false
(3 rows)
[ec2-user@ip-172-31-43-16 cockroach-v24.1.0.linux-amd64]$
```

The CockroachDB cluster was deployed on three AWS EC2 instances. A bank workload benchmark was executed against node1. During node failures, throughput temporarily dropped, sometimes to 0 ops/sec, and tail latency increased significantly. After quorum and leaseholders stabilized, the system resumed

progress. CockroachDB preserved consistency by either completing transactions safely, returning ambiguous result errors, or stopping progress when quorum was unavailable.

This demonstrates that CockroachDB ensures strong consistency even during node failures, at the cost of temporary unavailability while the system reconfigures.